General Biology 1

First Semester

Supplementary Materials

Module 10

The Cell Membrane

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What I know

Directions: Read each situation carefully and choose the best answer.

1. What is the main function of the cell membrane?

a. To produce energy
b. To control what enters and leaves the cell
c. To store genetic information
d. To create proteins

2. What does “selectively permeable” mean?

a. Allows all substances to pass through
b. Blocks all substances from entering
c. Allows some substances to pass while blocking others
d. Only allows water to pass

3. Which structure helps maintain homeostasis in the cell?

a. Nucleus
b. Ribosome
c. Cell membrane
d. Cytoplasm

4. How does the cell membrane help maintain balance inside the cell?

a. By producing ATP
b. By regulating the movement of substances
c. By breaking down food
d. By storing nutrients

5. What might happen if the cell membrane stops functioning properly?

a. The cell will divide faster
b. The cell will lose control of its internal environment
c. The cell will produce more proteins
d. The cell will grow larger

What I Need to Know

This supplementary module is designed to help you understand the **Cell Membrane**. In this lesson, you will explore how the selective permeability of the cell membrane protects and helps the cell to maintain homeostasis.

Lesson 10 The Cell Membrane

The cell membrane is a thin, flexible layer that surrounds the cell and separates it from its environment. It is made mainly of a phospholipid bilayer with embedded proteins that help regulate what enters and leaves the cell. This selective permeability allows the cell to maintain balance and proper functioning.

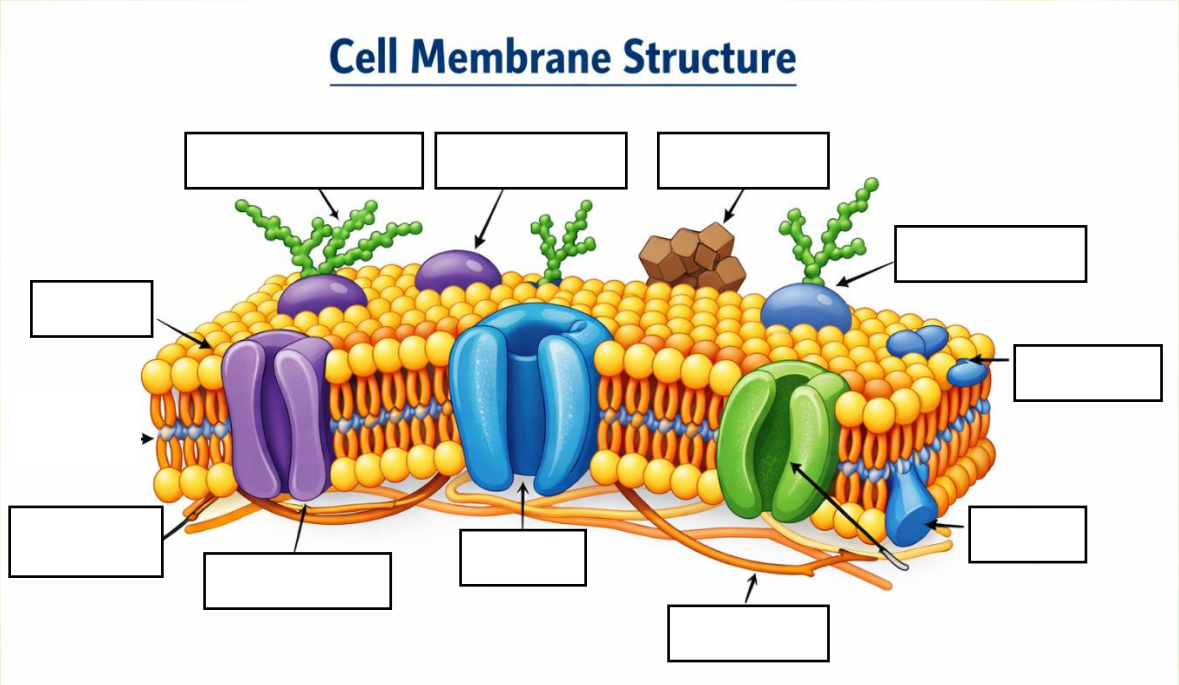
After going through this supplementary module, you are expected to:

1. describe the structure and function of the cell membrane, including its selective permeability;
2. explain how the cell membrane regulates the movement of substances in and out of the cell to maintain homeostasis; and
3. analyze the importance of the cell membrane in maintaining proper cellular conditions and supporting cell survival.

What's In

Activity 1: "The Cell's Shield: Discovering Membrane Structure"

Directions: Identify the parts of the cell membrane.



Hydrophobic Tails	Peripheral Protein	Glycoprotein
Channel Protein	Hydrophobic Heads	Lipid
Integral Protein	Carrier Protein	Glycolipid
		Cytoskeleton
		Cholesterol

The **cell membrane** is a flexible, dynamic structure that surrounds the cell and controls the movement of substances in and out. It follows the **fluid mosaic model**, meaning it is made of a phospholipid bilayer with proteins and other molecules embedded within it.

The main structure is the **phospholipid bilayer**, composed of hydrophilic (water-loving) heads and hydrophobic (water-fearing) tails. The **hydrophilic heads** face outward toward the watery environments inside and outside the cell, while the **hydrophobic tails** face inward, forming a barrier that prevents most substances from freely passing through. These phospholipids are a type of lipid, which gives the membrane its flexibility and structure.

Embedded within this bilayer are different types of proteins that perform specific functions. **Integral proteins** span the entire membrane and are involved in transport and communication. Among them are **channel proteins**, which form pores that allow certain molecules or ions to pass through, and carrier proteins, which bind to substances and change shape to move them across the membrane. **Peripheral proteins** are found on the surface of the membrane and often help with support, signaling, or maintaining the cell's shape.

The membrane also contains **glycoproteins** and **glycolipids**, which have carbohydrate chains attached. These structures are important for cell recognition, communication, and adhesion, helping cells identify each other and interact properly.

Another important component is **cholesterol**, which is found between the phospholipids. Cholesterol helps maintain the membrane's **fluidity and stability**, preventing it from becoming too rigid in cold conditions or too fluid in warm conditions.

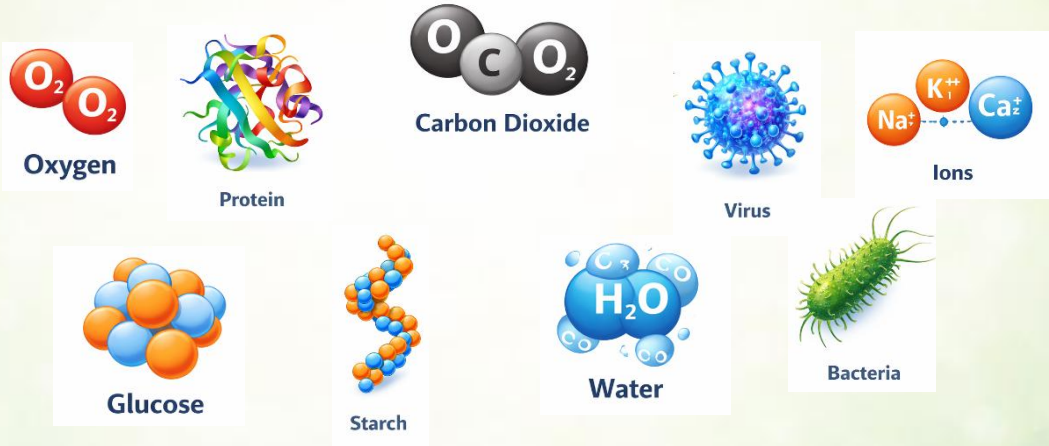
Finally, the **cytoskeleton** is a network of protein fibers attached to the inner side of the membrane. It provides structural support, helps maintain cell shape, and assists in movement and organization within the cell.

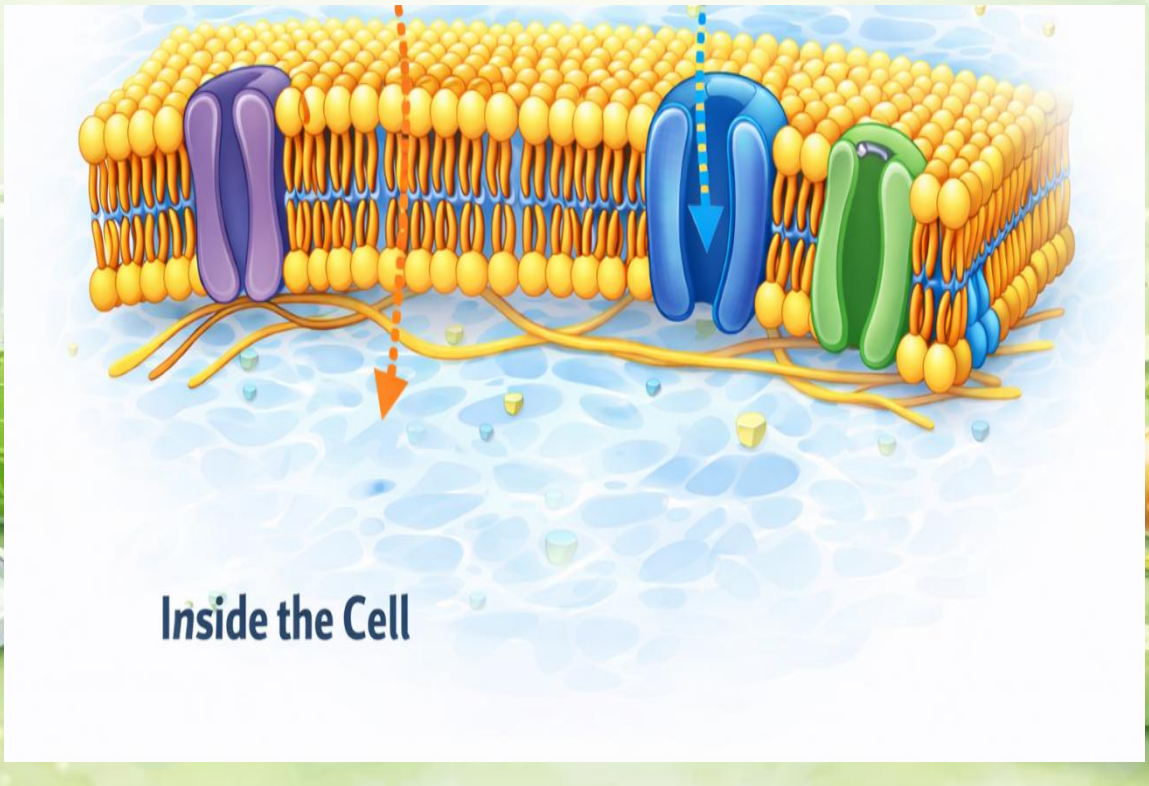
Overall, the structure of the cell membrane allows it to function as a selectively permeable barrier, protecting the cell while regulating the transport of substances necessary for survival.

What's New

Activity 2: "Choose What Enters: The Selective Cell Membrane"

Directions: Choose the substances that the cell membrane allows to enter the cell. Put it inside the cell.





Guide Questions:

1. What does it mean that the cell membrane is selectively permeable?

   _____

   _____

2. Why are some substances blocked to cross the membrane?

   _____

   _____

3. Why are some substances blocked or require assistance to cross the membrane?

   _____

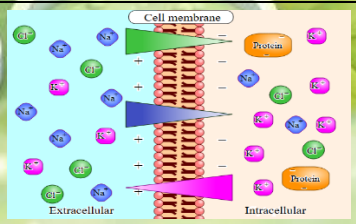
   _____

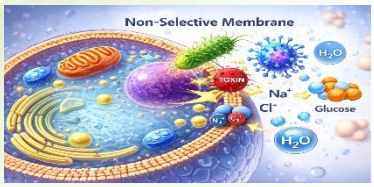
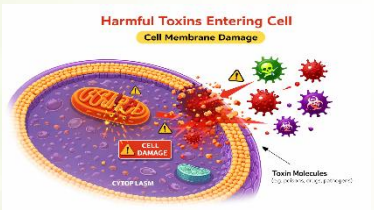
The **cell membrane** is a selectively permeable barrier that regulates the movement of substances in and out of the cell to maintain **homeostasis**. It does not allow all substances to enter because uncontrolled entry could disrupt the cell's internal balance, allow harmful materials inside, or cause the cell to swell or shrink. Essential substances such as **oxygen** enter the cell for energy production, **water** moves in and out to maintain balance, **glucose** provides energy, and **ions** help regulate important processes like nerve signaling and muscle contraction, while **carbon dioxide** exits as a waste product. In contrast, **large molecules** like **proteins** and **starch**, as well as harmful substances such as **toxins** and **pathogens**, are restricted or blocked. By carefully selecting what enters and leaves, the cell membrane ensures that the cell remains stable, functional, and able to survive.

What's More

Activity 3: "Cell in Crisis: The Membrane Matters!"

Directions: Analyze the given illustration and answer what will happen to the cell if these scenarios can be uncontrolled by the cell membrane.

Illustration	Scenario	What will happen to the cell?
	Too many ions enter the cell	

	Glucose cannot enter the cell	
	Membrane becomes non-selective	
	Oxygen cannot enter the cell	
	Harmful toxins enter freely	

Guide Questions:

1. Which scenario shows the most immediate danger to the cell? Why?

   _____

   _____

2. How do these scenarios show the importance of selective permeability?

   _____

   _____

To experience **MORE** interactivity in this module, scan this QR code.

What I've Learned

Directions: Encircle the letter of the correct answer.

1. A cell membrane becomes damaged and allows all substances to pass through freely. What is the most likely outcome?
   a. The cell will function normally
   b. The cell will produce more energy
   c. The cell will lose internal balance and may die
   d. The cell will grow faster
2. A scientist places two identical cells in different solutions. Cell A is placed in a solution with a high concentration of salt, while Cell B is placed in pure water. After some time, Cell A shrinks while Cell B swells. Which explanation best describes what is happening at the cell membrane level?
   a. The cell membrane actively pumps water into Cell A and out of Cell B
   b. Water moves across the membrane by osmosis, from low solute concentration to high solute concentration
   c. The cell membrane allows salt to freely enter both cells, causing size changes
   d. Water moves equally in both directions, so there is no net movementA
   3. A cell membrane becomes more permeable to ions without using protein channels. What does this suggest about the membrane structure?
   a. The hydrophobic core is damaged
   b. The membrane has more cholesterol
   c. The proteins are working better
   d. The phospholipids are more organized
   4. A cell is exposed to a toxin that blocks all membrane protein channels but does not damage the phospholipid bilayer. Which process will be most directly affected?
   a. Diffusion of oxygen into the cell
   b. Movement of water across the membrane
   c. Facilitated diffusion of glucose into the cell
   d. Passage of small nonpolar molecules through the membrane5. Which scenario best shows the importance of selective permeability in maintaining homeostasis?
   a. A cell allows all molecules to enter freely
   b. A cell blocks all substances from entering
   c. A cell regulates entry and exit based on its needs
   d. A cell stops all transport processes